

Krishnakumar Valliappan

Software Engineer | Toronto, Canada

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SUMMARY

Software Engineer with 3+ years building end-to-end products across responsive web interfaces, APIs, AI workflows, and data systems using TypeScript, Vue, React, Python, FastAPI, Express, and SQL.

EXPERIENCE

Full-Stack Developer — IPAC Consulting Canada · Jan 2025 – Present

- Built and shipped an **AI-guided pricing assistant** with FastAPI, Vue 3, SQLAlchemy, SSE, and structured LLM outputs, supporting 100+ sessions while reducing reliance on manual customer contact.
- Built a **tenant-scoped document Q&A workflow** using OpenAI file-search RAG, SSE streaming, company and site filters, and page-linked citations for uploaded PDFs.
- Shipped an **embeddable iframe widget** that gave external-site visitors responsive access to the assistant while preserving session context across visits.
- Built a **full-stack image annotation system** with Vue, Leaflet, Express, and Prisma, replacing a workflow that cross-referenced spreadsheets and image files.
- Built an **interactive spatial data map** that overlays live operational records on annotated layouts with filters, clustered markers, and drill-down details.
- Created a **PDF.js and HTML Canvas viewer** with lazy rendering for large documents and role-gated download/print controls, including blocked print shortcuts for restricted files.
- Reduced repeat API calls by **~30%** on affected screens using localStorage caching with two-minute TTLs, namespaced keys, and mutation-triggered invalidation.
- Built an **idempotent data ingestion pipeline** that normalized external spreadsheet data into a relational quote cache and protected against stale-record overwrites.
- Implemented **draft autosave and lifecycle workflows** with debounced persistence, flush-on-exit, and server-authoritative transitions to prevent data loss and invalid state changes.
- Containerized FastAPI and Vue services with **Docker Compose, Nginx, and Caddy**; added Bitbucket checks for migrations, health tests, and frontend types before manual Coolify deployments.

Junior Developer — IA Flow Elements India · Jan 2021 – Jul 2022

- Automated recurring **AWS data synchronization with Python**, removing manual execution from ~90% of scheduled runs.
- Built **React dashboards** with filtering and comparison views, giving stakeholders one interface for operational data analysis.
- Rebuilt internal interfaces with **Tailwind CSS** to improve mobile responsiveness and standardize spacing, typography, and component styling.
- Resolved production issues across internal tools by tracing data and UI defects, refining React behavior, and shipping fixes with business stakeholders.

PROJECTS

JobScout — github.com/krishnavalliappan/JobScout Python · Selenium · OpenAI · Notion API

- Built an open-source **job research and application tracking tool** with 45 GitHub stars that extracts structured job data, drafts application materials with OpenAI, and syncs tracking with Notion.

EDUCATION

Master of Engineering — Concordia University, Montreal 2022 – 2024

SKILLS

Languages	Python, TypeScript, JavaScript, SQL
Backend	FastAPI, Express.js, Node.js, REST APIs, Server-Sent Events, Pydantic, Prisma, SQLAlchemy
Frontend	Vue 3, React, Next.js, Pinia, Tailwind CSS, Leaflet, HTML5 Canvas
AI	OpenAI APIs, LLM workflow orchestration, function calling, structured outputs, RAG
Data & Cloud	PostgreSQL, MySQL, SQLite, Docker, AWS, CI/CD, Nginx, Git